Q7(a)
A box sliding on a rough surface slows down because friction retards its motion.

A person walking on rough ground has a foot pushing the ground backwards. By Newton's Third Law, the ground pushes the foot forward. The force responsible for the push is frictional in nature.

Q7(b)
For vertical equilbrium of horse, $N = mg$ where N is the normal force .

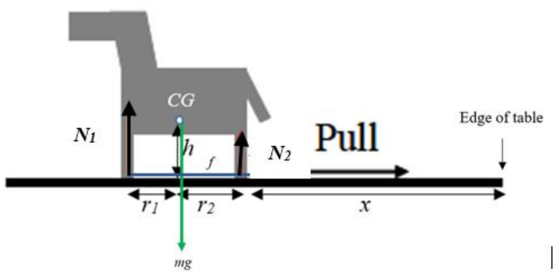
$$\mu = \frac{f}{N} \Rightarrow f = 0.750 \text{ x } 0.100 \text{ x } 9.81 = 0.7358$$
$$\approx 0.736 \text{ N}$$

Q7(c)(i)
N2L, $f = ma$

$$a = \frac{0.7358}{0.100} = \mathbf{7.36 \text{ m s}^{-2}}$$

Q7(c)(ii)
Using “$v^2 = u^2 + 2as$"
We have $v = \sqrt{2ax} = \sqrt{2(7.358)(0.300)}$
$\approx$ **2.10 m s^{-1}**

Q7(d)(i)



Q7(d)(ii)1.

$$\boldsymbol{N_1 + N_2 = mg}$$

Q7(d)(ii)2.

$$\boldsymbol{N_1 r_1 = N_2 r_2 + fh}$$

Q7(d)(iii).
$(mg - N_2)r_1 = N_2 r_2 + fh$

$N_2 = \frac{mgr_1 - fh}{r_1 + r_2} = \frac{mgr_1 - \mu mgh}{r_1 + r_2}$

$\therefore N_2 = mg\frac{r_1 - \mu h}{r_1 + r_2}$

Q7(d)(iv)
$N_2 = (0.100)(9.81)\frac{0.0500 - 0.750(0.0500)}{0.150}$
$\approx$ **0.0818 N**

$N_1 = mg - N_2 = (0.100)(9.81) - 0.0818$ $\approx$ **0.899 N**

Q7(d)(v)
If the back legs of the horse lose contact with thetable, $N_2 = 0$.
$\therefore N_2 = mg\frac{r_1 - \mu h}{r_1 + r_2} = 0$ or $r_1 - \mu h = 0$
Hence $h = \frac{r_1}{\mu} = \frac{0.0500}{0.750} \approx$ **0.0667 m**

Q7(d)(vi)
Contact is maintained if $N_2 > 0$ or

$r_1 - \mu h > 0$. Hence if $r_1 > \mu h$, the back legs of the horse will not lose contact with the table, i.e. have a big value for r_1.

Secondly, we can have a small value for μ i.e. the horse is pulled on a smooth cloth.
CLT topic removed on account of COVID-19.

******** END ********